Introduction to Machine Learning

This concise module is designed for AI-agent based learning and quick revision.

Introduction

Machine Learning (ML) enables computers to learn patterns from data and make predictions without being explicitly programmed.

Applications

Used in healthcare, finance, recommendation systems, fraud detection, autonomous vehicles, NLP and computer vision.

Types of Learning

Supervised: learns from labeled data. Unsupervised: finds hidden patterns. Reinforcement: learns by rewards and penalties.

Classification

Predicts discrete classes such as spam/not spam.

Regression

Predicts continuous values such as house prices.

Overfitting

Model memorizes training data and performs poorly on new data.

Underfitting

Model is too simple and fails to capture patterns.

Model Evaluation

Common metrics include Accuracy, Precision, Recall, F1-score and Cross Validation.

Quick Revision: Revise definitions first, understand key concepts, then practice MCQs.